FUNCTORS ON QCOH COME FROM MORPHISMS (AFFINE DIAGONAL CASE)

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4847-4854 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4856-4930, the complete author proof (73 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_stacks-stacks-more-morphisms.json (pb=2782, stmt 4847-4854, proof 4856-4930).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: top-square commutativity and the functor correspondence 'details omitted' - named-mechanism bookkeeping; uniqueness rides on the fully-faithful lemma.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-affine: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4575–4583. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-easy-kunneth: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4738–4747. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-flat: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4716–4721. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-fully-faithful: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4792–4799. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universally-injective: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4646–4650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma proposition-stack-fpqc: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4321–4325. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0783: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4930–4930. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04YC: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4930–4930. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be a Noetherian algebraic stack with affine diagonal. Let B be a ring. Let $F : QCoh(\mathcal{O}_\mathcal{X}) \to Mod_B$ be a right exact tensor functor which commutes with direct sums. Then F comes from a unique morphism* $\mathrm{Spec}(B) \to \mathcal{X}$.

Proof. Choose a surjective smooth morphism $g : U \to \mathcal{X}$ with $U = \mathrm{Spec}(A)$ affine, see Properties of Stacks, Lemma 04YC. Apply Lemma lemma-affine to get the finite type commutative B-algebra $C = F(g_{QCoh,*}\mathcal{O}_U)$ and the ring map $A \to C$. By Lemma lemma-universally-injective the ring map $B \to C$ is universally injective. Consider the algebra

$$C \otimes_B C = F(g_{QCoh,*}\mathcal{O}_U \otimes_{\mathcal{O}_\mathcal{X}} g_{QCoh,*}\mathcal{O}_U)$$

Since g is flat, quasi-compact, and quasi-separated by Lemma lemma-easy-kunneth we have the first equality in

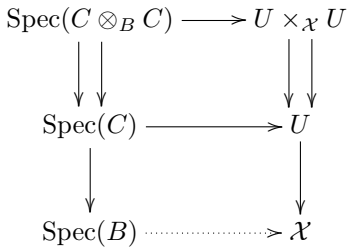
$$g_{QCoh,*}\mathcal{O}_U \otimes_{\mathcal{O}_\mathcal{X}} g_{QCoh,*}\mathcal{O}_U = f_{QCoh,*}\mathcal{O}_{U\times_\mathcal{X} U} = g_{QCoh,*}(\mathrm{pr}_{2,*}\mathcal{O}_{U\times_\mathcal{X} U})$$

where $f : U \times_\mathcal{X} U \to \mathcal{X}$ is the obvious morphism and $\mathrm{pr}_2 : U \times_\mathcal{X} U \to U$ is the second projection. The second equality follows from Cohomology of Stacks, Lemma 0783 and $f = g \circ \mathrm{pr}_2$. Since the diagonal of $\mathcal{X}$ is affine, we see that $U \times_\mathcal{X} U = \mathrm{Spec}(R)$ is affine. Let us use $\mathrm{pr}_2 : A \to R$ to view R as an A-algebra. All in all we obtain

$$C \otimes_B C = F(g_{QCoh,*}\mathcal{O}_U \otimes_{\mathcal{O}_\mathcal{X}} g_{QCoh,*}\mathcal{O}_U) = F(g_{QCoh,*}(\mathrm{pr}_{2,*}\mathcal{O}_{U\times_\mathcal{X} U})) = R \otimes_A C$$

where the final equality follows from the final statement of Lemma lemma-affine. Since $A \to R$ is flat (because pr_2 is flat as a base change of $U \to \mathcal{X}$), we conclude that $C \otimes_B C$ is flat over C. By Lemma lemma-flat we conclude that $B \to C$ is faithfully flat.

We claim there is a solid commutative diagram



The arrow $\mathrm{Spec}(C) \to U = \mathrm{Spec}(A)$ comes from the ring map $A \to C$ in the statement of Lemma lemma-affine. The arrow $\mathrm{Spec}(C \otimes_B C) \to U \times_{\mathcal{X}} U$ similarly comes from the ring map $R \to C \otimes_B C$. To verify the top square commutes use Lemma lemma-fully-faithful; details omitted. We conclude we get the dotted arrow $\mathrm{Spec}(B) \to \mathcal{X}$ by Proposition proposition-stack-fpqc.

The statement that F is the functor corresponding to pullback by the dotted arrow is also clear from this and the corresponding statement in Lemma lemma-affine. Details omitted. □